

Milestone 2

Objective.

In this stage, the focus is on examining how disaster declarations change over time. The objective goes beyond simply creating visualizations—it aims to uncover meaningful patterns, long-term trends, and seasonal variations within the data.

As outlined in the project workflow for visualizing US natural disasters, this phase specifically emphasizes:

- Analyzing temporal trends over different time periods
- Comparing trends across various types of disaster incidents

Validate Time Data.

This step prepares the dataset for time-based analysis by converting the *Declaration Date* into datetime format and extracting year and month features. The column is also renamed for consistency, and missing date values are checked to ensure data quality.

Yearly Trend Analysis.

This code performs yearly trend analysis by using `groupby('year').size()` to count the total number of disaster declarations per year and `groupby(['year', 'Incident type']).size().unstack()` to compare trends across different incident types. It is used to identify long-term patterns and changes over time in disaster occurrences. A line graph is created using `plt.plot()` to visualize how disasters increase or decrease each year.

Output: The graph shows yearly disaster trends, and it is observed that between 2000 and 2010, the number of disasters is the highest, indicating a peak period of disaster declarations.

Incident Type Trends Over Time.

This code visualizes incident type trends over time by plotting each disaster category separately using a loop (`for column in type_trends.columns`). It uses `plt.plot()` to show how the number of disasters for each incident type changes year by year, allowing easy comparison between different disaster types. This method helps identify which disasters are increasing or decreasing over time.

Output: The graph shows that floods and storms occur more frequently compared to other disaster types, and some categories display a rising trend over the years, possibly due to seasonal patterns and increasing extreme weather events.

Seasonality Analysis.

This code performs seasonality analysis by grouping the data using `groupby('month').size()` to count disasters in each month. A bar chart is created using `plt.bar()` to visualize how disaster frequency varies across all 12 months. This method is used to identify monthly patterns and seasonal variations in disaster occurrences.

Output: The graph shows that some months have higher disaster counts, indicating clear seasonality. Peaks are usually seen due to hurricane seasons, monsoon periods, winter storms, and summer wildfires, which increase disaster frequency in specific months.

Advanced Time Insights.

This code provides advanced time-based insights by analyzing long-term trends and growth patterns in disaster data. The `rolling(window=3).mean()` function is used to calculate a 3-year moving average, which smooths short-term fluctuations and highlights the overall trend. Additionally, `pct_change()` is used to compute the year-over-year growth rate, showing how disaster counts increase or decrease annually. These methods help in understanding both trend stability and variability over time.

Output: The moving average graph shows a gradual increasing trend in disasters with reduced fluctuations, while the growth rate graph highlights irregular spikes and drops, indicating that some years experience sudden increases due to extreme events like hurricanes or wildfires.

This code identifies the top 5 peak disaster years by sorting the yearly disaster counts in descending order using `sort_values()` and selecting the top values with `head(5)`. It is used to highlight the years with the highest disaster occurrences for focused analysis.

Output: The result shows that 2005 has the highest number of disasters (4736), followed by 2011, 2008, 2004, and 1993. These years stand out with significantly higher counts, possibly due to major events like hurricanes, floods, or wildfires.

CONCLUSION.

The temporal analysis of disaster data reveals clear trends, patterns, and seasonality over time. The number of disaster declarations shows an overall increasing trend, especially between 2000 and 2010, indicating a rise in disaster occurrences. Certain disaster types like floods and storms are more frequent and show noticeable growth. Monthly analysis highlights seasonal patterns, with specific months experiencing higher disaster counts due to climatic conditions. Advanced insights, such as moving averages and growth rates, confirm long-term increases with short-term fluctuations. Additionally, peak years like 2005 stand out due to extreme natural events.

Overall, the analysis suggests that climate factors, extreme weather events, and improved reporting systems play a significant role in the rising number of disasters.